

Tile and Slide : A New Framework for Scaling NeRF from Local to Global 3D Earth Observation

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Objective

Scaling up the NeRF [4] method for large satellite imagery at high resolution with the following constraints:

1. The algorithm should run in linear time (or faster) with respect to the input size: $\text{time} = \mathcal{O}(N)$
2. The runtime memory should never exceed a fixed memory budget: $\text{memory} = \mathcal{O}(1)$
3. The tiled algorithm A_s and the reference (untiled) algorithm A should produce comparable quality, if not identical, results on any input x : $A_s(x) \approx A(x)$

3D Tile generation

Individual non-overlapping AABBs (Axis-Aligned Bounding Boxes) are defined in ground coordinates, ensuring contiguous and efficient spatial segmentation for processing.

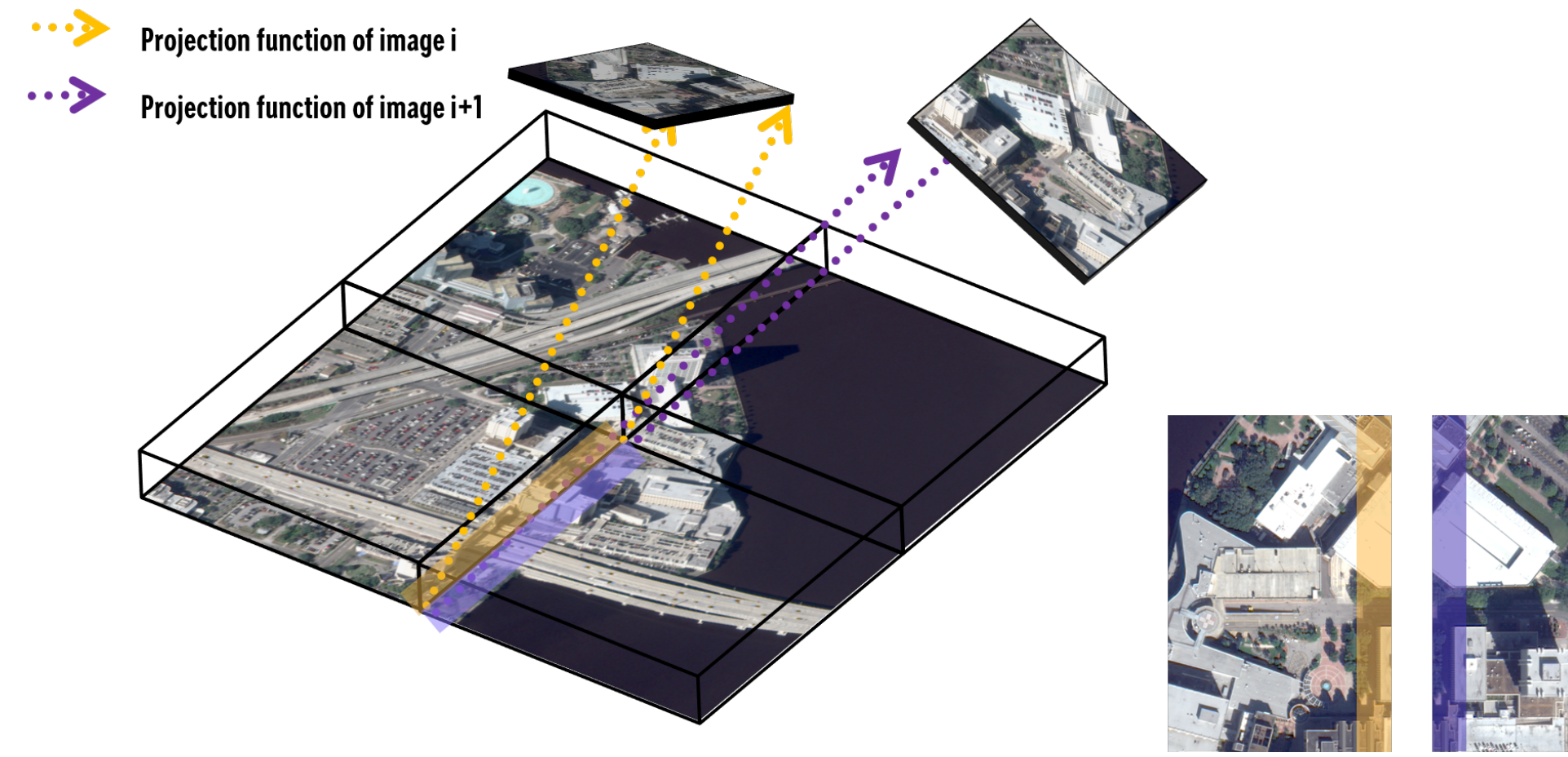
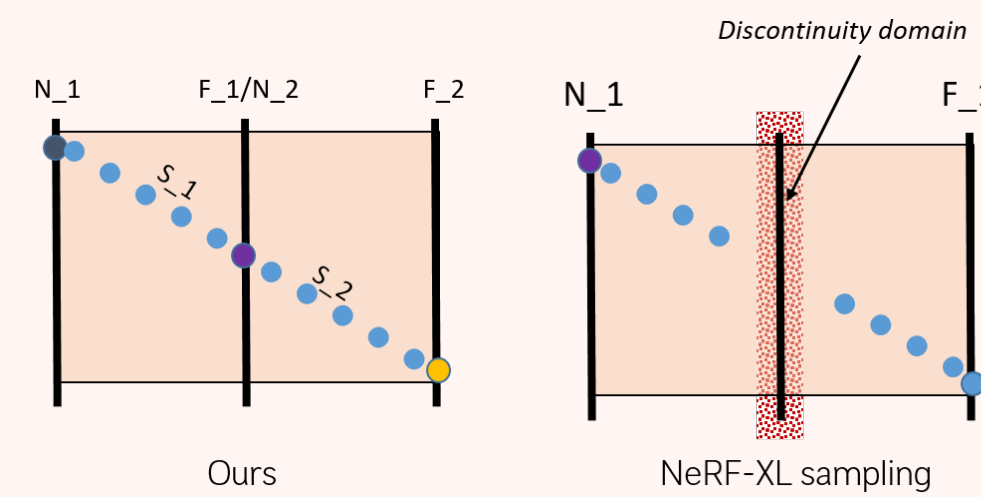


Figure 1. Image partitioning is based on ground coverage, taking into account the minimum and maximum altitudes of the AABBs. The areas highlighted in orange and purple indicate regions of the image with overlap.

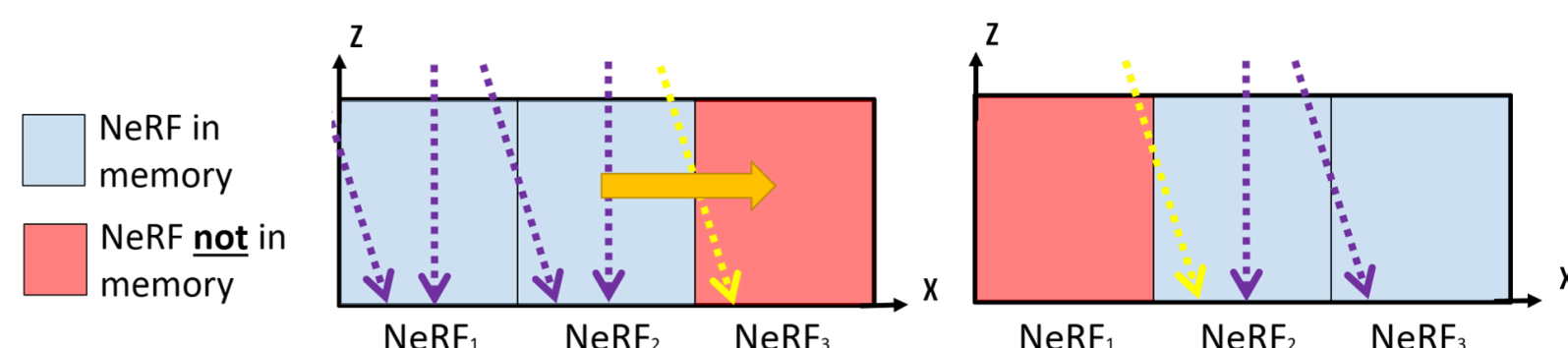
Segmented sampler

Segmentation of rays before sampling (left) ensures domain continuity, unlike a method where a ray is sampled and then distributed according to its membership in the AABB (right).

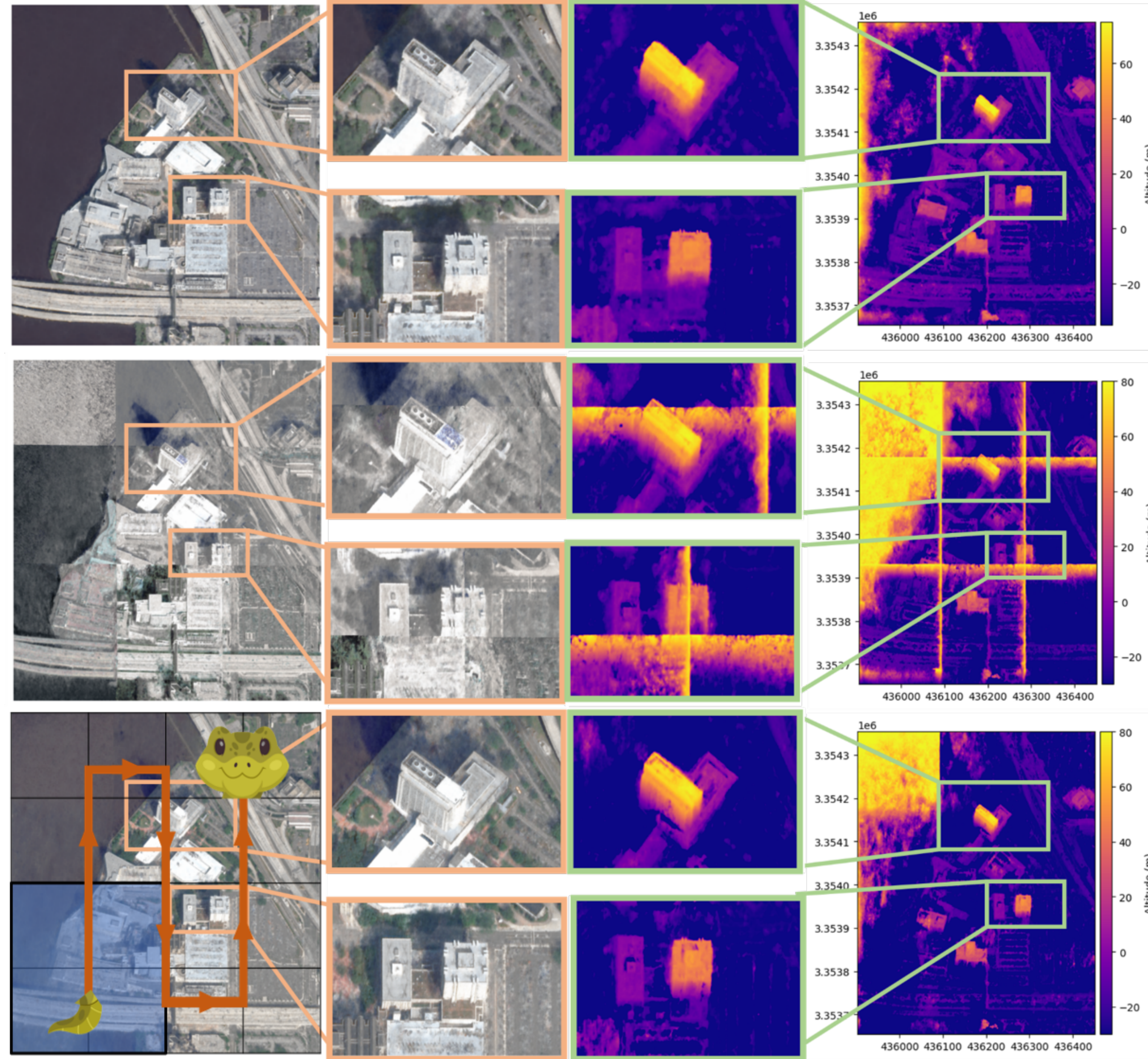


Sliding window

Only rays intersecting the bounding box currently in memory or its neighboring regions (purple arrows), which are also present in memory, will be accepted for sampling. Conversely, rays intersecting a NeRF that is not currently in memory will be discarded (yellow arrows).



Paper results on DFC [5] 2019



- The reference algorithm (**Top**) utilizes a single model to learn the entire scene. Errors are caused by shadows and transient effects.
- The unscaled case (**Middle**), where a single NeRF is trained at a time, demonstrates significant artifacts along the 3D tile edges.
- **Snake-NeRF (Bottom)**: Uses a 2×2 sliding window across the scene in a "snake" pattern. It shows performance similar to the reference algorithm while being applicable to large areas.

Quantitative results

Scene	# NeRFs	Dataset	Area (km^2)	PSNR	Time	GPU Memory
Reference case	1	JAX 214	0.28	19.10	45min	15GB
Unscaled	9	JAX 214	0.28	15.82	45min	8GB
Ours (window 2×2)	9	JAX 214	0.28	19.30	45min	15GB
Ours (window 2×2)	36	Paris	3.20	19.05	6h	15GB

Latest Results on Pleiades NEO Images

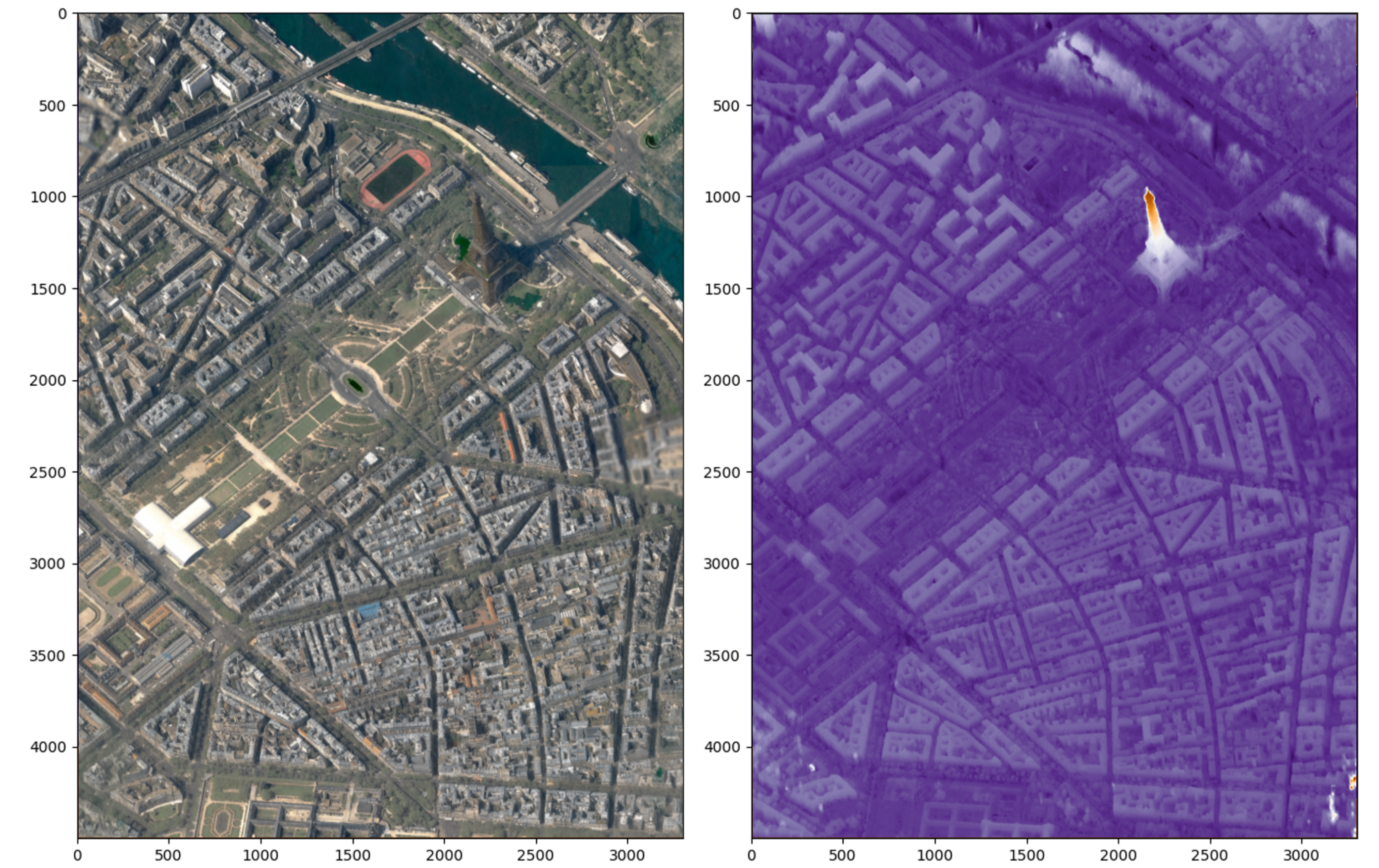


Figure 2. Novel view synthesis (left) and depth map (right) from Champ de Mars (Tour Eiffel) in Paris.

Conclusion

Snake-NeRF is a consistent approach that trains 4 sub-NeRFs at a time using a 2×2 sliding window. The window is slid across the scene to cover the entire area in a "snake" pattern.

- Maintains strong spatial locality by reducing the distance between grid areas to be processed and minimizing the need to reload recently processed data.
- Ensures optimal use of GPU memory, regardless of its capacity.

References

- [1] Camille Billouard, Dawa Derksen, Alexandre Constantin, and Bruno Vallet. Tile and slide: A new framework for scaling nerf from local to global 3d earth observation. *arXiv preprint arXiv:2507.01631*, 2025.
- [2] Camille Billouard, Dawa Derksen, Emmanuelle Sarrazin, and Bruno Vallet. SAT-NGP: Unleashing neural graphics primitives for fast relightable transient-free 3D reconstruction from satellite imagery. In *IGARSS 2024-2024 IEEE International Geoscience and Remote Sensing Symposium*, pages 8749–8753. IEEE, 2024.
- [3] Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. *Communications of the ACM*, 65(1):99–106, 2021.
- [4] Thomas Müller, Alex Evans, Christoph Schied, and Alexander Keller. Instant neural graphics primitives with a multiresolution hash encoding. *ACM Trans. Graph.*, 41(4), 2022.
- [5] Bertrand Le Saux, Naoto Yokoya, Ronny Hänsch, and Myron Brown. Data fusion contest 2019 (dfc2019), 2019.